

COLLEGE OF ENGINEERING

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SIGNAL AND SYSTEM: ANALYSIS AND APPLICATIONS

REPORT FOR FOURIER TRANSFORM GROUP PROJECT

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1. INTRODUCTION

1.1 Project Objectives

- To explore the Fourier transform experimentally and learn its uses in modulation, demodulation, and signal processing.
- To gain hands-on experience with amplitude modulation and audio signals using MATLAB and Simulink
- To examine a two-stage modulation and demodulation system using filtering as a case study
- To address theoretical issues pertaining to the ideas of the Fourier transform and its spectral consequences.

1.2 Theoretical Background

Analysis in the frequency domain is made possible by the Fourier Transform, which breaks down signals into their component frequencies. Understanding time-domain signals and their spectrum properties is a common application in signal processing. The process of altering a high-frequency carrier signal's amplitude in response to a modulating signal is known as amplitude modulation (AM). Demodulation is the reverse of the modulation process to recover the original signal. In order to improve signal quality and lower noise, two-stage modulation and demodulation add several modulation stages, sometimes requiring filtering.

This process is often used to reduce interference and improve signal quality, especially in environments with a lot of interference or high-frequency signals. Filtering, such as the use of low- or high-pass filters, also plays an important role in removing unwanted frequency components, ensuring that the transmitted or received signal is clear and free from interference. This approach can be used to realise a variety of practical applications, including in telecommunications, audio systems, and signal engineering.

2. METHODOLOGY

2.1 Tools and Software Used

- MATLAB
  Load and visualise the signal visualisation in the frequency domain. The Fourier transform uses “fft” to transform computation. Generate the signal based coding and simulink to perform amplitude modulation (AM) signal and demodulated signal.

- Hardware
  Microphone for recording voice signal

2.2 Steps in Simulation/Experiment

Steps to perform simulations and experiments for the Fourier Transform Project. First, for Task 1, audio signals such as bird sound, instrumentals, and dialects will be converted from the time domain to the frequency domain. In Task 2, one's voice will be converted from the time domain to the frequency domain. The audio file will be downloaded and placed in the MATLAB directory before being visualised in the time domain using the `audioread` function. Then, the Fourier Transform will be performed using the `fft` function, and the frequency spectrum of the signal will be plotted to compare its frequency components.

Next, for Task 3, an amplitude modulation (AM) simulation will be performed using MATLAB SIMULINK. A sinusoidal modulated signal will be generated along with a carrier signal with a higher frequency. The modulation process will be performed by multiplying both signals to produce an AM signal. The time domain and frequency spectrum of all these signals will be visualised for analysis.

In addition, for Task 4, one's voice will be recorded using the `audiorecorder` function in MATLAB and analysed using the Fourier Transform. The amplitude modulation process will be applied to the voice signal, and this signal will then go through a demodulation process to recover the original signal. All time domain and frequency spectra of the original signal, carrier, AM signal, and demodulated signal will

be plotted and analysed. Finally, for task 5 a case study was given to show the Fourier Transform of the equation given. And sketch the frequency spectrum to perform a system in Fourier Transform with filtering.

2.3 Explanation of the MATLAB Code Used

2.3.1 Task 1: Fourier Transform of Audio Signals

```
s = audioread('C:\Users\User\OneDrive\Documents\UITM\DEGREE
UITM\SEM 5\ESE562\MINI PROJECT\Sound little birds singing in
the trees.wav');
s = s(:, 1); % Extract the first (left) audio channel.
plot(s); % Plot the audio signal in the time domain.
xlabel('Time (s)'); % Label the x-axis as time.
ylabel('Amplitude'); % Label the y-axis as amplitude.
title('Time Domain Signal: Bird Sound');% Add a title
grid on; % Enable the grid for better visualization.
```

The code above is for generating a time-domain (Bird Sound) signal, this code loads an audio file located in a specific directory using the 'audioread' function, where the audio signal is stored in the 's' variable. Then, the first channel (left channel) of audio is extracted for analysis using 's(:,1)'.

```
s = audioread('C:\Users\User\OneDrive\Documents\UITM\DEGREE
UITM\SEM 5\ESE562\MINI PROJECT\Sound guitar stroke up
slow.wav');
s = s(:, 1);
plot(s);
xlabel('Time (s)');
ylabel('Amplitude');
title('Time Domain Signal: instrument Sound');
grid on;
```

The code above is for generating a time-domain (Instrument Sound) signal, this signal is plotted in the time domain using ‘plot’ function, with the x-axis labeled as “ Time(s) “ to indicate time, and the y-axis labeled as “Amplitude” to indicate the amplitude of the signal. The signal title is given as “ Time domain Signal: instrument Sound”. Finally, the grid is activated using ‘grid on’ to facilitate analysis of the signal on the signal.

```
s = audioread('C:\Users\User\OneDrive\Documents\UITM\DEGREE
UITM\SEM 5\ESE562\MINI PROJECT\Sound good morning.mp3');
s = s(:, 1);
plot(s);
xlabel('Time (s)');
ylabel('Amplitude');
title('Time Domain Signal: Dialect Sound');
grid on;
```

The code above is for generating a time-domain (Dialect Sound) signal, and the code used is the same as the code to generate the time-domain signal for bird sound and instrument sound.

```
s1 =audioread('C:\Users\User\OneDrive\Documents\UITM\DEGREE
UITM\SEM 5\ESE562\MINI PROJECT\Sound little birds singing in
the trees.wav');
Fs = 44100; % Set the sampling frequency.
N1 = length(s1);  % Get the number of samples in the signal.
transform1 = fft(s1,N1); % Compute the Fourier Transform.
magTransform1 = abs(transform1);
faxis1 = ((-0.5:1/N1:0.5-1/N1)*Fs).';
plot(faxis1, fftshift(magTransform1(:,1)),'m-+');
hold on % Allow additional plots on the same figure.
xlim([-2000 2000]); % Limit the frequency axis to -2000 to 2000
Hz.
```

```
lgd = legend('Bird Sound'); % Add a legend to the plot.
lgd.FontSize = 10;
set(gca,'fontsize',10)
grid on
xlabel('Frequency (Hz)');
ylabel('Spectrum Magnitude');
title('Frequency Domain Signal: Bird Sound');
```

The code above is for generating a frequency domain (Bird Sound) signal, this code analyses a bird sound signal in the frequency domain. `Fs` sets the sampling rate (44100 Hz), and `N1` counts the number of samples in the `s1` signal.

```
s2 =audioread('C:\Users\User\OneDrive\Documents\UITM\DEGREE
UITM\SEM 5\ESE562\MINI PROJECT\Sound guitar stroke up
slow.wav');
Fs = 44100;
N2 = length(s2);
transform2 = fft(s2,N2);
magTransform2 = abs(transform2);
faxis2 = ((-0.5:1/N2:0.5-1/N2)*Fs).';
hold on
plot(faxis2, fftshift(magTransform2(:,1)),'b-s');
xlim([-2000 2000]);
lgd = legend('Instrument Sound');
lgd.FontSize = 10;
set(gca,'fontsize',10)
grid on
xlabel('Frequency (Hz)');
ylabel('Spectrum Magnitude');
title('Frequency Domain Signal: Instrument Sound');
```

The code above is for generating a frequency domain (Instrument Sound) signal, a Fourier transform is performed with ‘fft’ function, producing a frequency spectrum of the

signals, and the magnitude of the spectrum is taken using ‘abs’. The ‘faxis1’ variable produces the frequency axis to be plotted.

```
s3 =audioread('C:\Users\User\OneDrive\Documents\UITM\DEGREE
UITM\SEM 5\ESE562\MINI PROJECT\Sound good morning.mp3');
Fs = 44100;
N3 = length(s3);
transform3 = fft(s3,N3);
magTransform3 = abs(transform3);
faxis3 = ((-0.5:1/N3:0.5-1/N3)*Fs).';
hold on
plot(faxis3, fftshift(magTransform3), 'k-p');
xlim([-2000 2000]);
lgd = legend('Dialect Sound');
lgd.FontSize = 10;
set(gca,'fontsize',10)
grid on
xlabel('Frequency (Hz)');
ylabel('Spectrum Magnitude');
title('Frequency Domain Signal: Dialect Sound');
```

The code above is for generating a frequency domain (Dialect Sound) signal, a Fourier transform is performed with the ‘fft’ function also used in this code. The frequency spectrum signal is plotted using ‘plot’, with frequencies limited between -2000 Hz and 2000 Hz by using ‘xlim’. X-axis and Y-axis labels, signal titles, and grids are added. For the title add “Frequency Domain Signal: Dialect Sound'.

2.3.2 Task 2: Fourier Transforms of Your Own Voice Signal

```
>> % Ensure your microphone is turned on and connected to
your computer.
% Record your voice for 3-5 seconds
fs = 44100; % Sampling frequency
nBits = 16; % Bits per sample
nChannels = 1; % Mono audio
duration = 5; % Duration of recording in seconds
```

Firstly, we start the code by setting the sampling frequency with 44100 Hz, Bit per sample is 16 and the duration is 5 seconds.

```
% Create an audiorecorder object
recObj = audiorecorder(fs, nBits, nChannels);
disp('Start speaking...');
recordblocking(recObj, duration);
disp('Recording complete.');
% Retrieve audio data
audioData = getaudiodata(recObj);
```

After that, we use the audiorecorder function in the program above to record our voices. Then, the program will give an indication to the user to start speaking and recording completely.

```
% Time-domain analysis
t = (0:length(audioData)-1) / fs; % Time vector
% Plot the time-domain signal
figure;
```

```
subplot(2, 1, 1); % Create a subplot for the time-domain
signal
plot(t, audioData, 'b');
xlabel('Time (s)');
ylabel('Amplitude');
title('Time Domain Signal: Own Voice');
grid on;
```

Then, the program will do a time-domain analysis and plot time-domain signals. For plotting time-domain signals, the x-axis is time (second) and the y-axis is amplitude.

```
% Frequency-domain analysis
N = length(audioData); % Length of the audio signal
transform = fft(audioData, N); % Fast Fourier Transform
magTransform = abs(transform); % Magnitude spectrum
faxis = (-0.5:1/N:0.5-1/N) * fs; % Frequency axis
% Plot the frequency-domain signal
subplot(2, 1, 2); % Create a subplot for the frequency-domain
signal
plot(faxis, fftshift(magTransform), 'm-+');
xlim([-2000 2000]); % Zoom into a relevant frequency range
xlabel('Frequency (Hz)');
ylabel('Spectrum Magnitude');
title('Frequency Domain Signal: Own Voice');
grid on;
% Add legends
legend('Own Voice (Frequency Domain)', 'Location', 'best');
set(gca, 'fontsize', 10);
```

Then, the program will do frequency-domain analysis and plot frequency-domain signals. For plotting time-domain signals, the x-axis is time (second) and the y-axis is amplitude.

2.3.3 Task 3: Amplitude Modulation (AM)

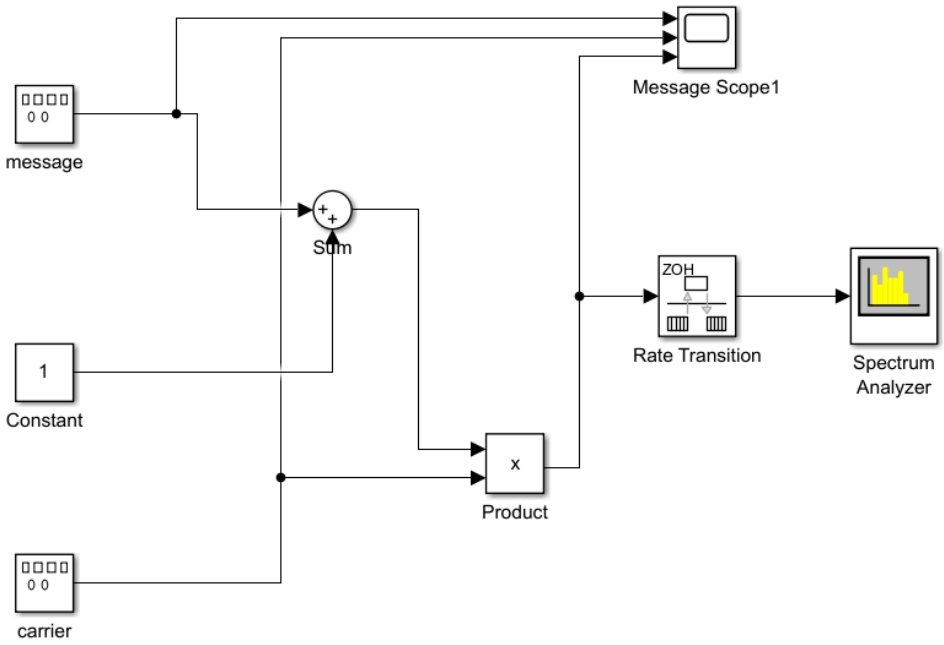


Diagram 2.3.3: Simulink block Amplitude Modulation (AM)

1. Open Simulink
    - Launch MATLAB
    - Open simulink in MATLAB. This will open the Simulink Library Browser
2. Create a new model
    - In the Simulink Library Browser, click on **File > New > Model** to create a new model
3. Add Block to the model
    - Scope: In the Simulink Library Browser, navigate to **Simulink > Sink** and drag the Scope block into your model
    - Rate transition: Go to **Simulink > Signal Routing** and drag the rate transition block into your model.
    - Spectrum Analyzer: Navigate to **DSP System Toolbox > Sink** and drag the Spectrum Analyzer block in your model
    - Constant: In the Library Browser, go to **Simulink > Sources** and drag the constant block into your model

- Product: Go to **Simulink > Math Operations** and drag the Product block into your model
- Sum: Go to **Simulink > Math Operations** and drag Sum block into to your model
- Carrier and Message: Go to **Simulink > Source** and drag the Signal Generator block into your model and rename the block as carrier and message

4. Connect the block like the diagram 2.3.3
5. Configure block parameter
    - Message: In parameter, set waveform in sine wave, Time (t) use simulation time, Amplitude set as 1, and Frequency set as 1 in unit Hertz.
    - Constant: Constant value as 1
    - Carrier: In parameter, set waveform in a sine wave, Time (t) use simulation time, Amplitude set as 1 and Frequency set as 10 in unit Hertz.
    - Rate transition: Set 0.01 as the output port sample time
6. Run the simulation
    - Click on the run button in the simulation toolbar to start the simulation
    - Double click on the scope and spectrum analyzer blocks to view the simulation results
7. Save the model
    - Save your model by clicking on File > Save As and giving it a name

2.3.4 Task 4: Voice Signal Modulation and Demodulation

```
>> % Ensure your microphone is turned on and connected to your
computer.
% Record your voice for 3-5 seconds
fs = 44100; % Sampling frequency
nBits = 16; % Bits per sample
nChannels = 1; % Mono audio
duration = 5; % Duration of recording in seconds
```

Firstly, we start the code by setting the sampling frequency with 44100 Hz, Bit per sample is 16 and the duration is 5 seconds.

```
% Create an audiorecorder object
recObj = audiorecorder(fs, nBits, nChannels);
disp('Start speaking...');
recordblocking(recObj, duration);
disp('Recording complete.');
% Retrieve audio data
audioData = getaudiodata(recObj);
```

After that, we use the audiorecorder function in the program above to record our voices. Then, the program will give an indication to the user to start speaking and recording completely.

```
% Time-domain analysis of the original signal
t = (0:length(audioData)-1) / fs; % Time vector
N = length(audioData); % Length of the audio signal
faxis = (-0.5:1/N:0.5-1/N) * fs; % Frequency axis

% Carrier signal
fc = 20; % Carrier frequency (20 Hz)
```

```
carrier = cos(2*pi*fc*t)'; % Carrier signal
```

Then, the program will do a time-domain analysis of the original signal and generate a carrier signal. For the carrier signal, we set the shape of the signal as cos and the frequency is 20 Hz.

```
% Amplitude Modulation
Ac = 1; % Carrier amplitude
normalizedVoice = audioData / max(abs(audioData)); % Normalize
audio signal
AM_signal = (Ac + normalizedVoice) .* carrier; % AM signal
% Demodulation using Hilbert Transform (Envelope Detection)
demodulatedSignal = abs(hilbert(AM_signal)); % Demodulated
signal
```

Besides that, the carrier amplitude we set as 1. The original voice signal is taken and modulated using amplitude modulation (AM) with a low-frequency (20 Hz) carrier signal and the demodulation process is performed using the Hilbert Transform to recover the original signal.

```
% Frequency analysis (Fourier Transform)
fftVoice = abs(fft(audioData, N)); % FFT of original signal
fftAM = abs(fft(AM_signal, N)); % FFT of AM signal
% Plot all results
Figure;
% Original Voice Signal (Time Domain)
subplot(3, 2, 1);
plot(t, audioData, 'b');
xlabel('Time (s)');
ylabel('Amplitude');
title('Original Voice Signal (Time Domain)');
grid on;
```

```
% Carrier Signal (Time Domain)
subplot(3, 2, 2);
plot(t, carrier, 'r');
xlabel('Time (s)');
ylabel('Amplitude');
title('Carrier Signal (Time Domain)');
grid on;
% AM Signal (Time Domain)
subplot(3, 2, 3);
plot(t, AM_signal, 'g');
xlabel('Time (s)');
ylabel('Amplitude');
title('AM Signal (Time Domain)');
grid on;
% Demodulated Signal (Time Domain)
subplot(3, 2, 4);
plot(t, demodulatedSignal - mean(demodulatedSignal), 'k');
xlabel('Time (s)');
ylabel('Amplitude');
title('Demodulated Signal (Time Domain)');
grid on;
```

Then, we proceed to frequency analysis in Fourier Transform and generate the time-domain signals (original voice signal, carrier signal, AM signal, and demodulated signal).

```
 % Original Voice Signal (Frequency Domain)
subplot(3, 2, 5);
plot(faxis, fftshift(fftVoice), 'm-+');
xlim([-2000 2000]); % Focus on relevant frequency range
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Frequency Spectrum of Original Voice Signal');
```

```
grid on;
% AM Signal (Frequency Domain)
subplot(3, 2, 6);
plot(faxis, fftshift(fftAM), 'c-+');
xlim([-12000 12000]); % Focus on relevant frequency range
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Frequency Spectrum of AM Signal');
grid on;
disp('All plots generated: Time-domain and Frequency-domain
analyses.');
```

Fourier transform is performed to see the frequency spectrum of the original signal and AM signal.

3. RESULTS AND ANALYSIS

3.1 Task 1: Fourier Transform of Audio Signals

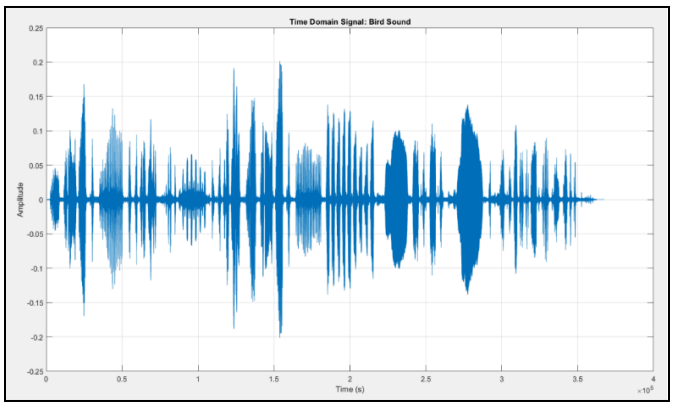


Figure 3.1.1: Time domain (Bird Sound)

Figure 3.1.1 shows the resulting plot shows the bird sound signal in the time domain, where the x-axis represents time and the y-axis represents amplitude. The visible waveform reflects the bird chirping sound, with high amplitudes indicating the louder parts of the sound, while low amplitudes represent the quieter parts or background noise. Chirping sounds typically produce short, sharp signals with rapid amplitude changes.

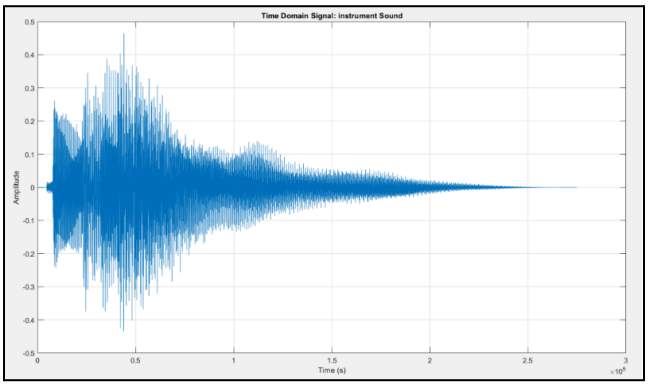


Figure 3.1.2: Time domain (Instrument Sound)

Figure 3.1.2 shows the dynamics and variations in sound strength throughout the recording, as well as the pattern of the guitar sound wave with amplitude oscillations. This pattern gives details about the guitar's acoustic properties, including rhythm, strum strength, and pitch length. And the amplitude decreases as the guitar sound slows down.

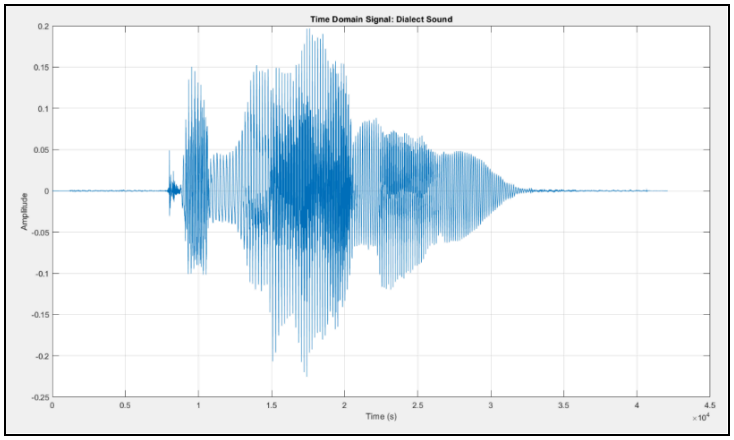


Figure 3.1.3: Time domain (Dialect Sound)

Figure 3.1.3 The results display the voice signal for "Good Morning" in the time domain. The amplitude variations on the signal reflect the pattern of the human voice when the phrase is performed. Changes in the waveform indicate characteristics including pitch, stress on pronunciation, and pauses between sounds.

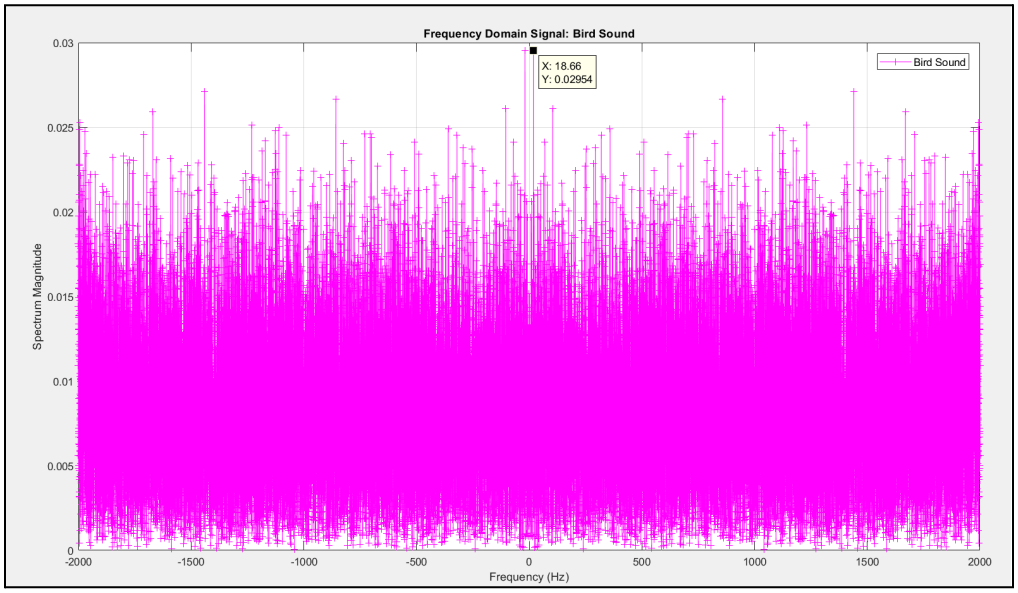


Figure 3.1.4: Frequency Domain (Bird Sound)

The resulting figure, 3.1.4 shows the frequency spectrum of bird sounds in the frequency domain. The y-axis displays the spectrum's magnitude, which indicates the intensity of the sound at each frequency, while the x-axis displays the frequency in Hertz (Hz). In the frequency range between -2000 Hz to 2000 Hz, peaks can be seen at certain frequencies

that indicate the main component of the bird chirping sound. These peaks show that the bird sound is made up of specific dominant frequencies that add to its uniqueness. Analysis is made easier by the grid and legend, and the frequency limitations (-2000 Hz to 2000 Hz) ensure that attention is focused on the important frequency range. The peak values at -18.66 Hz and 18.66 Hz with a peak magnitude of 0.02954 dB in the frequency domain of bird calls occur because the main frequency components in the bird call signal are around that range.

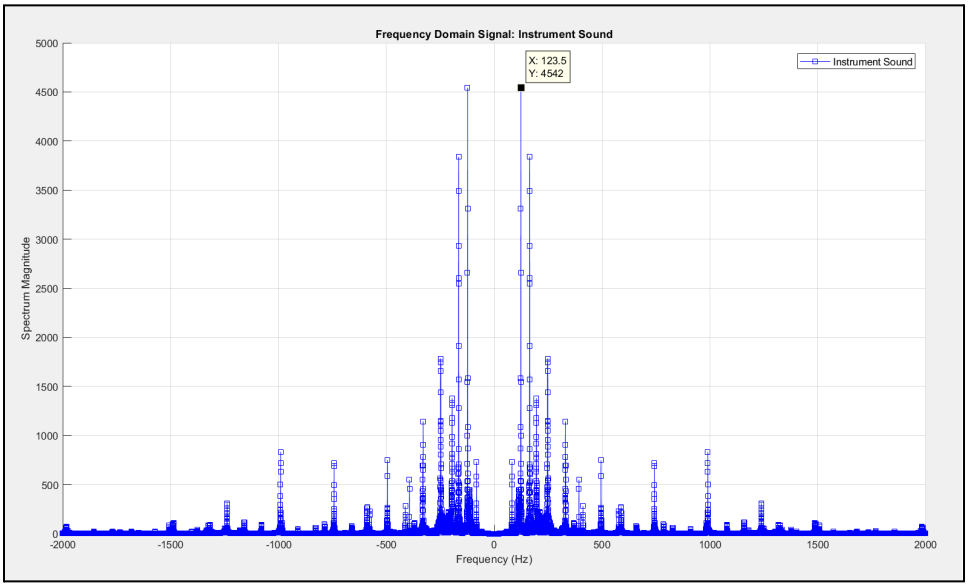


Figure 3.1.5: Frequency Domain (Instrument Sound)

Referring to Figure 3.1.5 in the frequency domain analysis of guitar stroke sounds using the Fourier transform, the resulting signal shows the signal magnitude spectrum in the frequency range. The prominent peaks are located at values around 123.5 Hz and 4542 Hz, which indicates the presence of a main frequency component in this range. This may be due to the nature of the guitar sound which produces low frequencies (such as the friction sound of the guitar strings) and high frequencies (such as harmonic sounds or overtones) which can be seen at higher frequencies. The peak magnitude values in this frequency range may be larger compared to bird sounds because guitar sounds are usually louder and clearer.

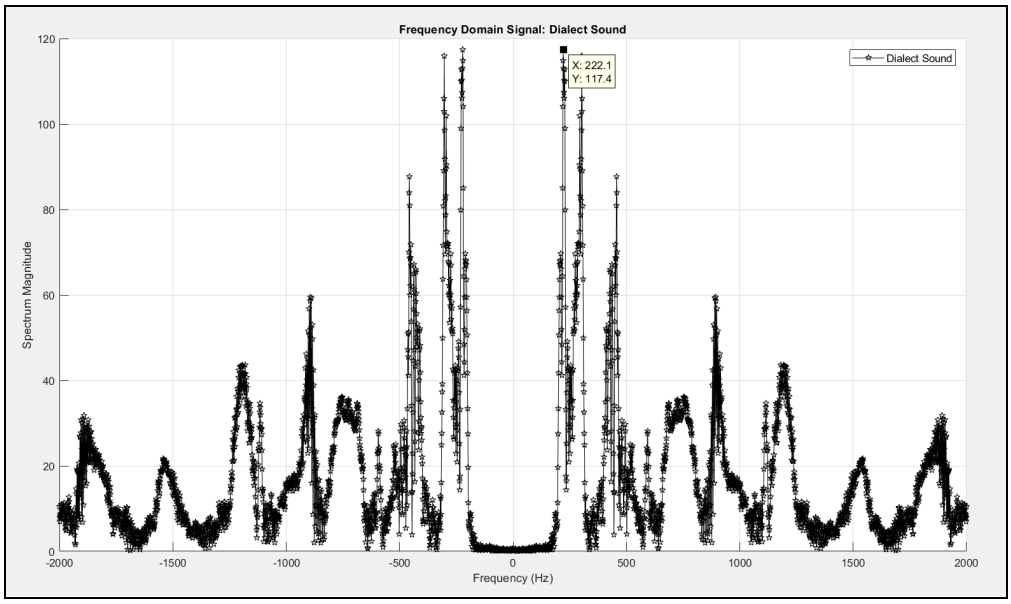


Figure 3.1.6: Frequency Domain (Dialect Sound)

Referring to Figure 3.1.6, in the frequency domain analysis of dialect sounds using the Fourier transform, the resulting signal shows the signal magnitude spectrum at a certain frequency range. The peaks on the signal are located at values of 222.1 Hz and -222.1 Hz on the x-axis, which indicates that the dominant frequency in this signal is around those values. The peak magnitude on the y-axis is 117.4, which indicates the strength of that frequency component in the dialect sound signal. This may refer to a certain fundamental or phonetic sound component in the human voice produced during conversation. Compared to bird or guitar sounds, dialect sounds show a more prominent frequency pattern at values around 222 Hz, which is commonly found in human speech or communication.

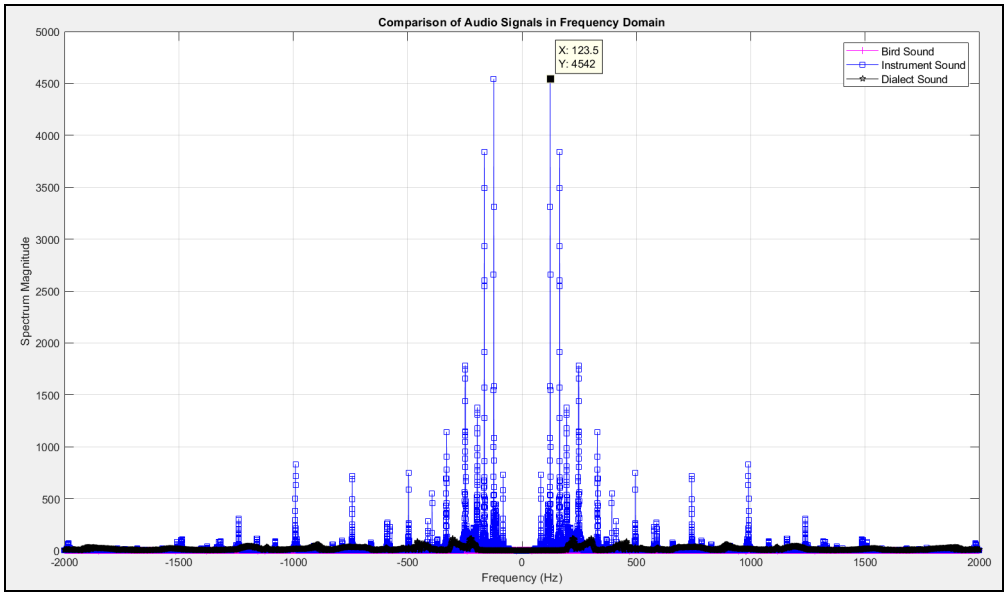


Figure 3.1.7: Comparison of Fourier Transforms for 3 Frequency Domain Signals

This task aims to explore the Fourier transform of three audio signals with durations ranging from 3 to 5 seconds. This process involves converting the signal from the time domain to the frequency domain and comparing the frequency components of each signal. Based on the signals displayed above, it can be seen that the frequency magnitudes of the three audio signals (Bird Sounds, Instrument Sounds, and Dialect Sounds) show significant differences. Bird Sounds (pink line) shows a wide frequency spread with lower magnitudes than the other two signals, reflecting the bird chirping sound pattern involving high and low frequencies. Musical Instrument Sounds (blue rectangle) are more dominant in the mid-frequency range near 0 Hz, with a sharp peak representing the periodic harmonic frequencies of the musical instrument. Dialect Sounds (black circles) show smaller peaks, indicating the low-frequency components closely related to the human voice.

When compared, Instrument Sounds show clearer and stronger peaks, reflecting the more dominant and periodic harmonies of musical instruments, compared to Dialect Sounds which have lower magnitude peaks because human speech sounds are less periodic. Bird Sounds, on the other hand, have a wider frequency spread and smaller peaks, indicating a more complex and natural sound pattern. To ensure accurate comparisons, all audio files were set to have the same duration between 3 and 5 seconds, using a uniform sampling frequency (Fs = 44100) for accurate frequency resolution, and normalizing the magnitude of each signal to focus on the relative differences between frequency components.

3.2 Task 2: Fourier Transforms of Your Own Voice Signal

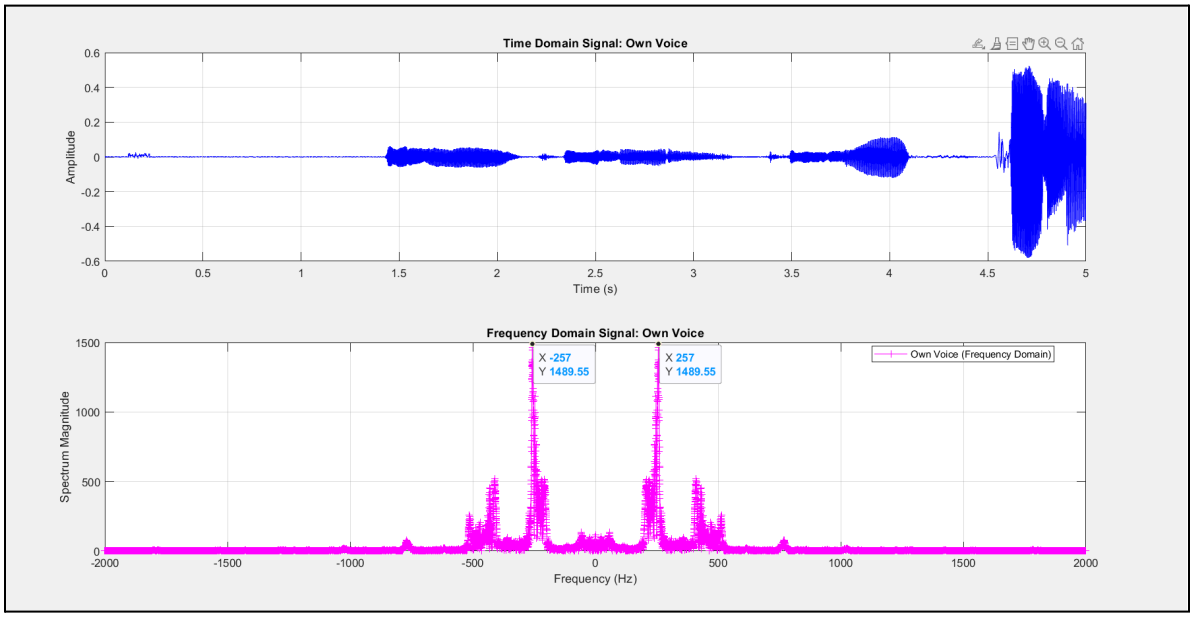


Figure 3.2.1: Own voice signal in time and frequency domain

A time-domain signal represents how the signal's amplitude varies over time. We use Fourier Transform to convert a time-domain signal into its frequency domain representation.Frequency-domain signal shows how the signal energy or power is distributed across various frequencies. We set -2000Hz and 2000 Hz as our frequency range. There is a peak at -257 Hz and 257 Hz indicating that the signal contains a strong component at this frequency. The peak magnitude is 1489.55 dB

3.3 Task 3: Amplitude Modulation (AM)

The AM signal is constructed by modulating the amplitude of a high frequency carrier wave (10 Hz) using a low frequency modulating signal (1 Hz). The modulating signal is a sine wave with a frequency of 1 Hz and amplitude of 1. The carrier signal is a sine wave with a frequency of 10 Hz. The amplitude of the carrier signal is 1.

AM Signal construction: s(t) = 1+ sin(2*pi*1*t)* sin(2*pi*10*t)

As the modulating signal varies between -1 and 1, it alters the amplitude of the carrier signal, creating the familiar envelope pattern of AM. The AM signal will have a peak at specific frequencies associated with the carrier (10 Hz) and the sidebands created by the modulation (1 Hz).

You will expect significant frequency components at:

fc-fm= 10 Hz - 1 Hz = 9 Hz

fc+fm=10 Hz + 1 Hz = 11 Hz

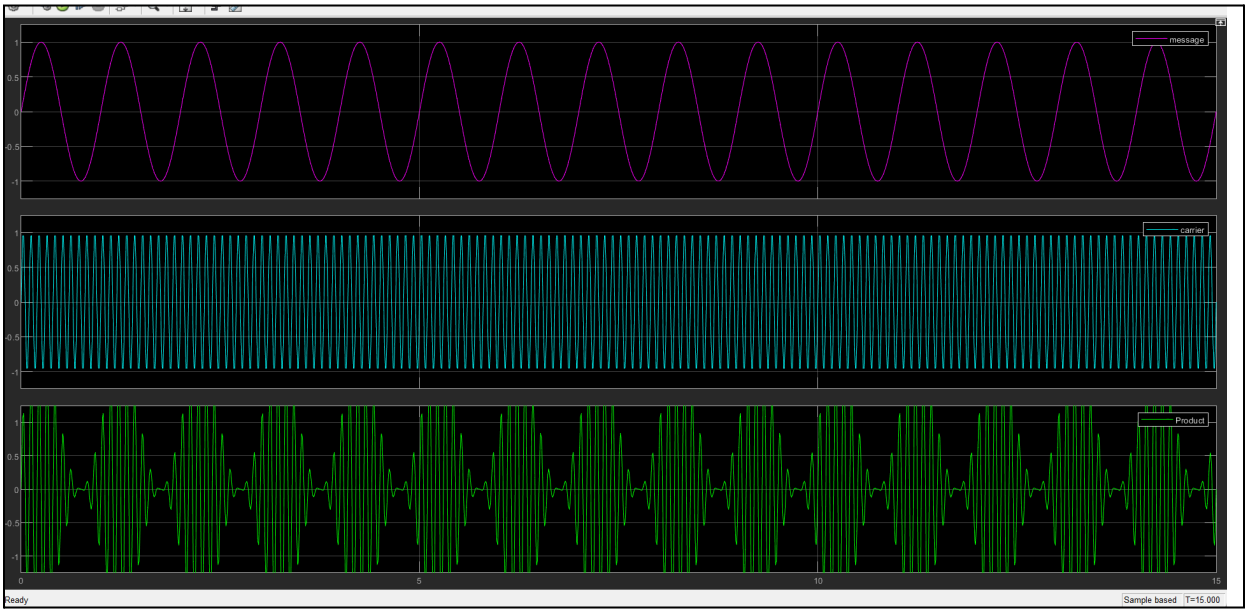


Figure 3.3.1: Plot time-domain waveform of the modulating signal, carrier signal, amplitude modulating (AM) signal

The modulating signal shows a slow oscillation and the carrier signal oscillates at a much higher frequency. The AM signal exhibiting the envelope formed by the modulating signal on the carrier frequencies

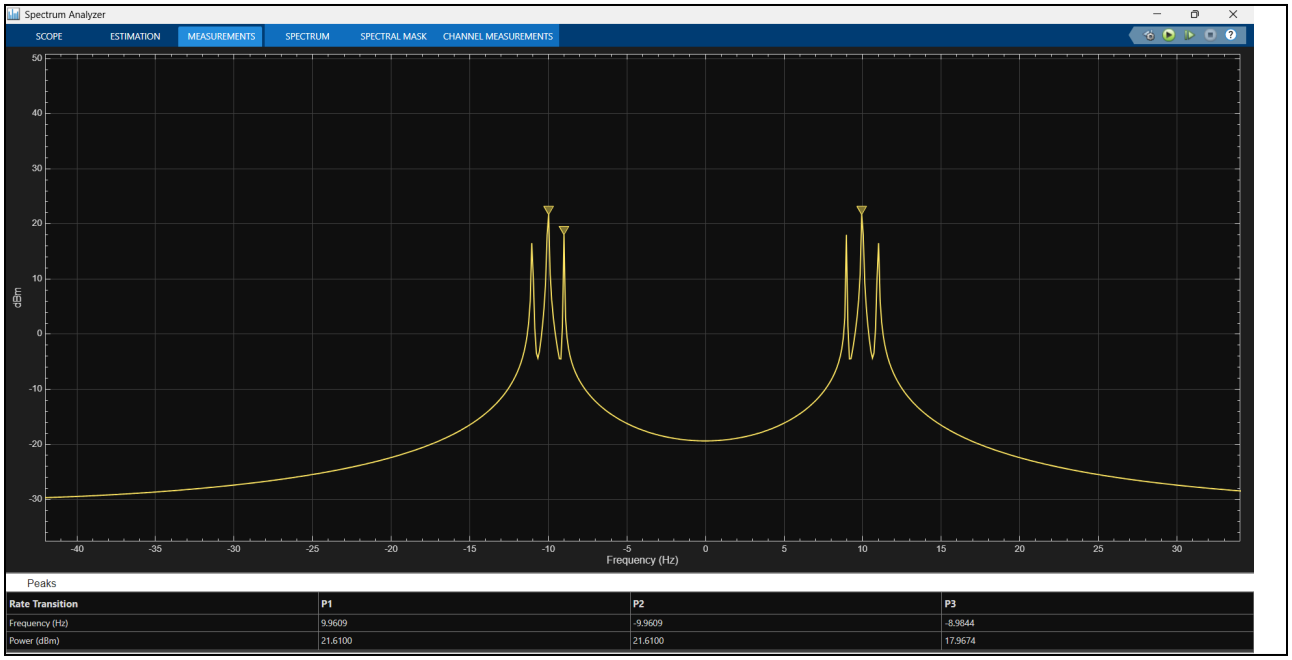


Figure 3.3.2: Plot the frequency spectrum of the Amplitude Modulating (AM) signal using the Fourier Transform

There is a peak at -9.609 Hz and 9.9609 Hz indicating that the signal contains a strong component at this frequency. The peak magnitude report (21.61 dB) is significant and indicates strong components at the extracted frequencies. The peak is approximately 9 Hz (lower sideband) and 11 Hz (upper sideband). A notable peak at the carrier frequency of 10 Hz and the sidebands confirming that the signal effectively carries information.

3.4 Task 4: Voice Signal Modulation and Demodulation

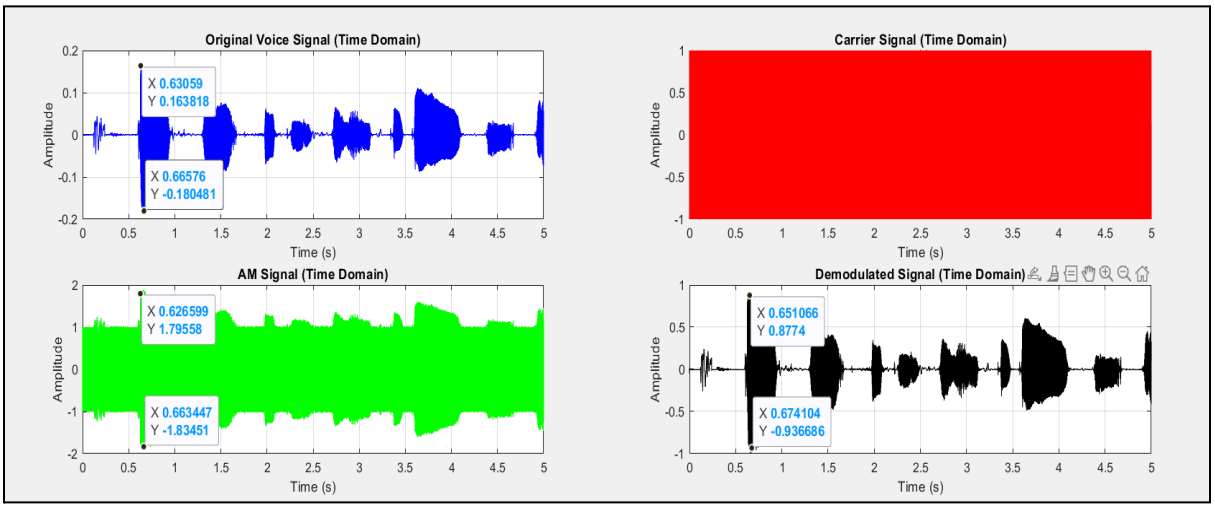


Figure 3.4.1:The time-domain waveforms of the original voice signal, carrier signal, AM signal, and demodulated signal.

In the Original Voice Signal, the peak time values are at 0.63059 and 0.66576 seconds, indicating two points where the original signal reaches its maximum and minimum values in the time domain. Meanwhile, the peak amplitude values are 0.163818 (positive) and -0.180481 (negative), reflecting the maximum and minimum oscillations of the original voice signal. In conclusion, this signal is the original voice signal without any modification, which is usually smaller in magnitude because it has not been modulated or amplified.

Then for the Amplitude Modulation Signal, the peak time values are 0.626599 and 0.663447, almost similar to the peak time in the original signal. This shows that the modulation does not change the peak time of the signal much. Meanwhile, the peak amplitude values are 1.79558 (positive) and -1.83451 (negative), much larger than the original amplitude. This change is due to the modulation process where the carrier signal increases its amplitude to ensure that the signal can be transmitted more efficiently. In conclusion, the AM signal combines the original signal with the carrier signal, resulting in a significant increase in amplitude but retaining the original time information.

In addition, the Demodulated Signal has peak time values of 0.651066 and 0.674104, which are slightly different from the peak time of the original signal. This

difference may be due to minor interference or inaccuracies in the demodulation process. Meanwhile, the peak amplitude values are 0.8774 (positive) and -0.936668 (negative), smaller than the amplitude of the AM signal but still larger than the amplitude of the original signal. This difference occurs because the Hilbert transform produces an envelope of the signal, which may not be completely uniform with the original signal. So in conclusion, the demodulation process attempts to re-extract the original information from the AM signal. Although the result is almost similar to the original signal, there are small differences in amplitude and time values due to artifacts from the modulation and demodulation processes.

In the meantime, the Carrier Signal plays an important role, The carrier signal is a high-frequency sinusoidal signal used in the modulation process. It increases the amplitude of the original signal, making it more suitable for long-distance transmission. The presence of the carrier signal explains why the amplitude of the AM signal is much higher than the original signal, which is 1.79558 and -1.83451, compared to 0.163818 and -0.180481 in the original signal.

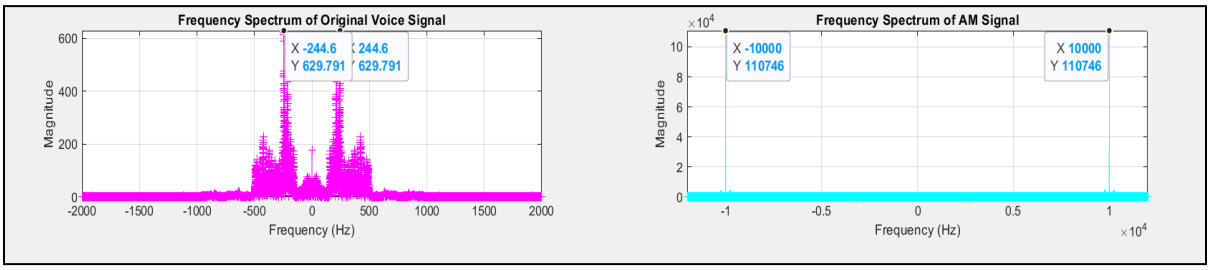


Figure 3.4.2: Frequency spectrum signals of the original voice signal and AM signal.

Referring to figure 3.4.2 on the Frequency spectrum of the original voice signal, the peak magnitude value is 629.791 in both positive and negative directions. While the peak frequency is -244.6 Hz (negative direction) and 244.6 Hz (positive direction). This peak magnitude value indicates the strength of the dominant frequency component in the original voice signal, while the peak frequency indicates the location of the dominant frequency. In this case, the main component of the signal lies at a low frequency of around 244.6 Hz, reflecting the characteristics of the original voice that is in the normal audio frequency range.

Then for the AM Signal frequency spectrum, the peak magnitude value is 110,746 in both positive and negative directions. While the peak frequency is -10,000 Hz (negative direction) and 10,000 Hz (positive direction). In an AM signal, the peak magnitude is much larger (110,746) due to the presence of the carrier frequency added during the modulation process. The peak frequencies are at plus or minus 10,000 Hz, which represents the main carrier frequency in this signal. This value is common for AM signals because the modulation process shifts the original frequency components to around the carrier frequency.

After studying the Hilbert transform, it was found that the Hilbert transform is a mathematical technique that transforms the original signal into a complex signal (analytic signal). This signal has two components, namely the real part (original signal) and the imaginary part (result of the Hilbert Transform of the original signal). This complex signal is used to analyse the frequency and phase properties of a signal. In the Hilbert Transform function itself, there is no explicit low-pass filtering. However, when the Hilbert Transform is used in the modulation or demodulation process, additional low-pass filtering is often used to remove unnecessary high-frequency components after demodulation. The Hilbert Transform produces an analytic signal by shifting the positive frequency phase by -90° and the negative frequency phase by +90°, and the DC (zero frequency) component is automatically eliminated.

In the amplitude modulation (AM) process, the Hilbert Transform is used to produce an envelope or package of the original signal. This envelope is an important component in preserving the original information of the signal. In demodulation, this envelope is re-extracted using the low-pass filtering process. If the carrier frequency is not too high, the envelope can be recovered correctly. When you use the Hilbert () function without specifying any additional parameters, it assumes that all frequency components of the signal can be transmitted. This function produces an analytic signal without using a cut-off frequency value. Technically, the Hilbert Transform itself is not a direct low-pass filter. It is a mathematical operation that transforms a real-time signal into an analytic signal by adding an imaginary part. However, low-pass filtering is often used in the Hilbert Transform process by adding.

3.5 Task 5: Case Study

1a) $F(w) = \frac{4}{3+jw}$ $\quad f(t) = 4e^{-3t}$

$$F(w) = \int_{-\infty}^{\infty} f(t)\, e^{-jwt}\, dt$$

i) $x(t) = e^{-j3t} f(t)$

$$x(t) = \mathcal{F}\{f(t)e^{-j3t}\} = \frac{4}{3+j(w+3)} \quad \text{(prop No 4)}$$

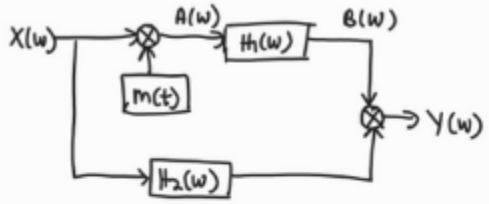
ii) $y(t) = f(t) \sin(4t)$

$$\mathcal{F}\{f(t)\} = \frac{1}{2j}\left[\mathcal{F}\{e^{j4t} - e^{-j4t}\}\right] f(t) \quad \text{(using prop. No 4)}$$

$$= \frac{1}{2j}\left[\frac{4}{3+j(w-4)} - \frac{4}{3+j(w+4)}\right]$$

$$y(t) = -2j(3+j(w-4)) + 2j(3+j(w+4))$$

b) Shows a system in Fourier Transform domain.



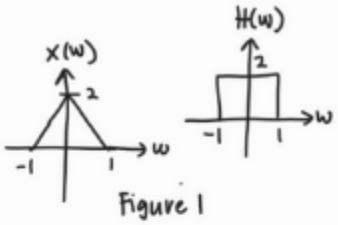
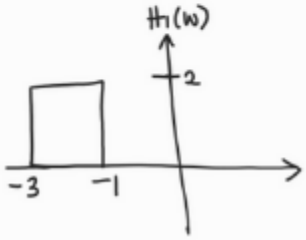


Figure 1

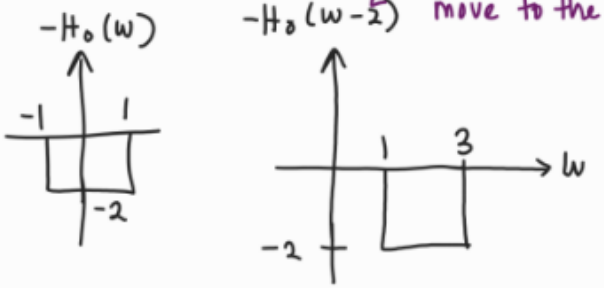
i) If $H_1(w) = [H_0(w+2)]$, sketch the frequency spectrum of $H_1(w)$

move to the left

$$H_1(w) = H_0(w+2) = H_0(w-(-2))$$



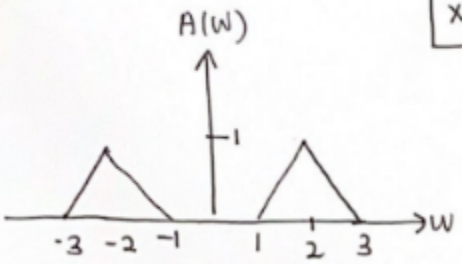
ii) IF $H_2(w) = -(H_0(w-2))$, sketch the frequency spectrum of $H_2(w)$

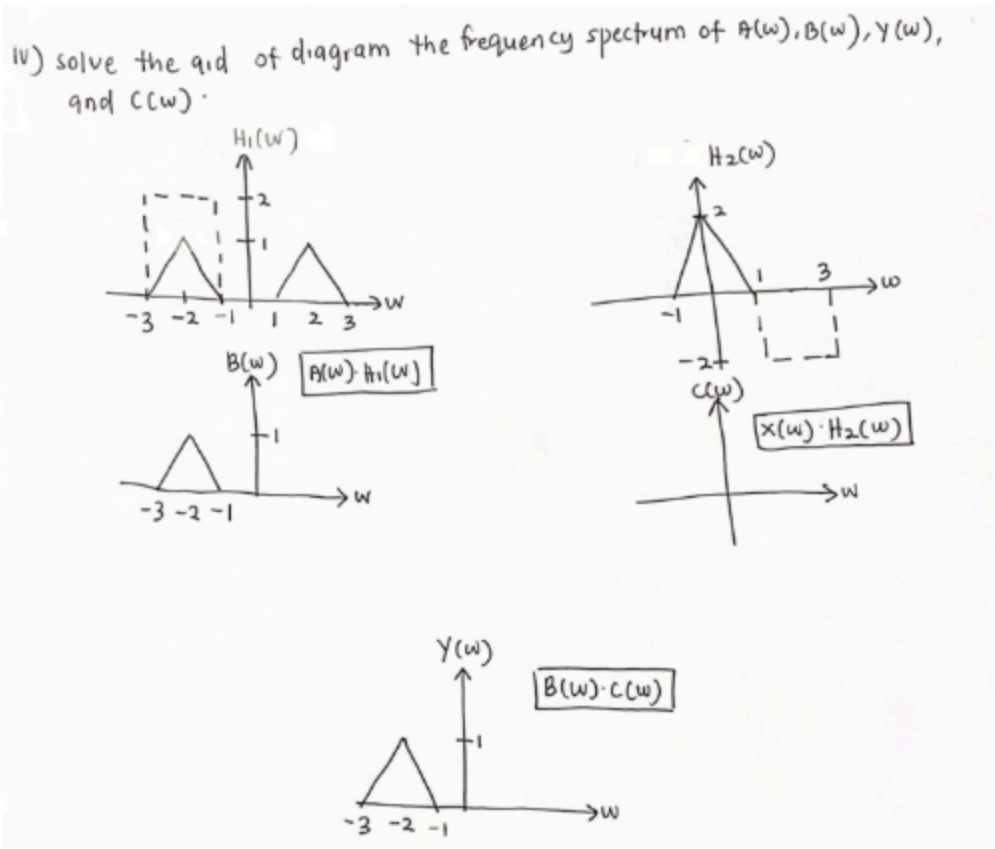


iii) suggest signal m(t) in order to produce A(w) as shown below:

$x(w) \cdot \cos 2wct$

$m(t) = \cos 2t$





For task 5 regarding the given case study, there are two questions, question a and question b. For question a, given x(t) = e^{-j3t}f(t), the Fourier Transform property shows that e^{-jωot} will cause the spectrum of F(ω)to shift to the right by ωo. Therefore, x(t) produces a spectrum shifted to ω+ 3. For y(t) = f(t)sin(4t), the sine can be rewritten as the combination of e^{j4t} and e^{-j4t} according to the property. This causes the spectrum of F(ω) to split into two parts, shifted to ω- 4 and ω + 4, which is the effect of frequency modulation. For (i) use property no 4 and for (ii) use Euler's Identity and property no 4.

In question b, for H1ω = Ho(ω+ 2), the spectrum (Hoω) is shifted to the left by 2 units, while H2(ω) = -Ho(ω - 2) is shifted to the right by 2 units with sign reversal on the magnitude. To generate A(ω), the signal m(t) can be determined based on the characteristics of A(ω) using modulation or other properties of the Fourier Transform. The spectra A(ω), B(ω), Y(ω), and C(ω) are each related through a multiplication operation with H1(ω) and H2(ω), which requires an understanding of the interaction between the frequency and time domains. Spectrum diagrams help visualise and determine changes in magnitude and frequency.

4. DISCUSSION AND INSIGHTS

4.1 Key Observations

Fourier transform effectively separates time domain signals into their frequency components. Amplitude modulation (AM) and demodulation accurately transmit and recover voice signals with proper carrier frequencies. Filtering enhances signal quality by reducing noise but may introduce phase distortion.

The main observations from the analysis conducted are that the Fourier transform successfully reveals unique frequency characteristics in audio signals, allowing a deeper understanding of the frequency components that exist in the signal. The amplitude modulation and demodulation (AM) process also show high reliability in the transmission and recovery of the original signal, proving the effectiveness of this technique in signal communication. In addition, the filtering performed in the two stage modulation/ demodulation system successfully improves signal quality by reducing unwanted noise, thus ensuring a cleaner and clearer signal. This observation shows the importance of the combination of frequency transformation, modulation, and filtering in audio signal optimization.

4.2 Challenges and Limitations

- Noise Interference

When a modulated signal is more open to noise in the transmission channel, the accuracy of signal recovery decreases. Signal quality can be greatly impacted by even little changes in the frequency spectrum, especially during the demodulation stage when the received signal may contain noise that affects the original data.

- Complexity of Computation

Calculating the Fourier Transform requires a large amount of computational power, especially when dealing with large data sets or signals that have high sampling rates. Real-time processing in MATLAB can be difficult to achieve if the computer hardware is underpowered or if MATLAB's performance is limited.

- Design of Filters

Because phase distortion and sharp cutoff must be balanced, designing the ideal filter is a difficult task. Improper filter design can cause signal distortion or reduce signal amplitude during the two-stage modulation or demodulation process, which will finally affect the signal's quality.

- Realistic Limitations

Physical limitations, such as microphone sensitivity, often affect the quality of a voice modulation recording signal. A poor-quality microphone may cause the signal to contain additional noise. Furthermore, incompatibility between software and hardware (for example, MATLAB and a microphone) may result in undesired distortions or inaccuracies in the signal.

4.3 Suggestions for Improvement and Future Directions

- Use of High-Quality Microphones

One improvement step is to use high-quality microphones that are more sensitive to the audio signal. This can enhance recording quality and reduce interference due to external noise. Better signal recovery will come from more accurate modulation and demodulation processes caused by a clearer signal input.

- Filter Optimization

To reduce amplitude reduction (attenuation) or signal distortion, filters might be altered. Users can obtain more effective noise removal without sacrificing quality by selecting a filter type, such as Butterworth or Chebyshev, and adjusting parameters, such as the cut-off frequency. To understand how the filter affects the signal, this can also be done with the MATLAB simulation tool.

- Using Algorithms for Real-Time Processing

Real-time processing techniques can be implemented to increase efficiency. For example, data can be processed more quickly with the use of technology like the Fast Fourier Transform (FFT). This is particularly helpful for applications that require fast processing, such as live voice transmission.

- Improved Hardware Integration

This project could be improved in the future to make use of specific hardware like microcontrollers designed for signal processing or Digital Signal Processors (DSP). Signal processing applications in real-world systems can be facilitated by this hardware, which can also speed up operations and lower system load. This makes the system suitable for advanced uses like voice recognition and wireless communication.

5. CONCLUSION

5.1 Summary of Key Findings

In this project, it has been found that the Fourier Transform is a very important tool in analysing and characterising signals. It makes it possible to convert time-domain data to frequency-domain data, helping in determining a signal's spectrum properties. In signal processing, this procedure is particularly helpful for identifying the most important frequencies in signal data and for evaluating audio signals.

In addition, this also shows the effectiveness of amplitude modulation (AM) in transmitting and receiving signals. Accurate delivery of information is possible by reducing signal interference through the use of appropriate modulation and demodulation techniques. In two stage modulation/demodulation systems, the focus is placed on the importance of using well designed filters to reduce noise interference and maintain the quality of the original signal. MATLAB software is a particularly helpful tool for solving technical issues and bringing complex signal processing systems into place because it makes it easier to simulate, visualise, and analyse signal processing tasks.

5.2 Impact and Applications of the Project

- Improving Telecommunications Systems' Efficiency

Basic telecommunications concepts like signal analysis, modulation, and demodulation are shown in this project. These ideas can be used to create communication systems that are more effective and adaptable to noise. For example, AM modulation can be used to ensure clear signal transmission even in highly interfered communication networks.

- Audio and Speech Processing

The techniques used in this project can be extended to audio quality improvement, music signal analysis, and voice recognition. Such as audio processing techniques are used by technology like virtual assistants like Siri and Alexa to understand spoken commands.

- Research and Education

Through this effort, researchers and students can gain a practical understanding of the signal processing idea. It enhances understanding of signal processing and the use of the Fourier Transform by applying the theory learned to use on an actual project. For example, students can understand how to analyse frequencies in signals such as bird calls or human speech.

- Applications of Engineering in the Real World

The project's output has applications in biological signal processing, including ECG (electrocardiogram) analysis, radar systems, and sonar technologies. In medical signal processing. For instance, the Fourier Transform is used to find patterns or abnormalities in cardiac signals, helping to generate more accurate diagnoses.

6. APPENDICES

4.2 Properties Of Fourier Transform

Operation	Time Function	Fourier Transform
Linearity	$af_1(t) + bf_2(t)$	$aF_1(\omega) + bF_2(\omega)$
Time shift	$f(t - t_0)$	$F(\omega)e^{-j\omega t_0}$
Time scaling	$f(at)$	$\frac{1}{\lvert a\rvert} F\left(\frac{\omega}{a}\right)$
Time transformation	$f(at - t_0)$	$\frac{1}{\lvert a\rvert} F\left(\frac{\omega}{a}\right)e^{-j\omega t_0/a}$
Duality	$F(t)$	$2\pi f(-\omega)$
Frequency shift	$f(t)e^{j\omega_0 t}$	$F(\omega-\omega_0)$
Convolution	$f_1(t)*f_2(t)$	$F_1(\omega)F_2(\omega)$
	$f_1(t)f_2(t)$	$\frac{1}{2\pi}F_1(\omega)*F_2(\omega)$
Differentiation	$\frac{d^n[f(t)]}{dt^n}$	$(j\omega)^n F(\omega)$
	$(-jt)^n f(t)$	$\frac{d^n[F(\omega)]}{d\omega^n}$
Integration	$\int_{-\infty}^{t} f(\tau)d\tau$	$\frac{1}{j\omega}F(\omega) + \pi F(0)\delta(\omega)$

7. REFERENCES

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[2] "Fourier Transform," Google Books, 2025. https://books.google.com.my/books?hl=en&lr=&id=j-KdDwAAQBAJ&oi=fnd&pg=PA291&dq=Hilbert+Transform+in+matlab&ots=0wqrqVjG1j&sig=to7npobuTDx5f5GHQ58aOWpwK8M&redir_esc=y#v=onepage&q=Hilbert%20Transform%20in%20matlab&f=false